

Smart MCQ Solver Challenge: DL & GenAI Project Report

Course: BSDA2001P: Intro to DL & GenAI Project

Topic: Multi-Option Question Answering & Ranking System

Primary Metric: Mean Average Precision at 3 (MAP@3)

1. Executive Summary

This project addresses the **Smart MCQ Solver Challenge**, where the primary objective is to evaluate machine learning and deep learning models capable of reasoning across complex multiple-choice prompts and ranking the top three most probable answers (A, B, C, D, E) using **MAP@3**. We experimented with three distinct modeling paradigms:

1. **Fine-Tuned BERT Transformer** (bert-base-uncased).
2. **Custom Deep Learning Model** (Bi-LSTM).
3. **Traditional Baseline** (TF-IDF + Cross-Validated Logistic Regression).

2. Introduction

- **Problem Statement:** The competition requires predicting ranked top-3 answer labels for multiple-choice questions containing a prompt and 5 choices (A–E).
- **Project Objective:** Build, fine-tune, and evaluate deep learning pipelines using PyTorch, Hugging Face, and Scikit-Learn. The goal is to maximize prediction accuracy and MAP@3 through proper tokenization, contextual embedding, and sequence modeling.

3. Dataset & Preprocessing

- **Dataset Description:** The dataset consists of structured MCQs containing:
 - id: Unique question identifier.
 - prompt: Context/Question stem.
 - A, B, C, D, E: The 5 option strings.
 - answer: truth label.
- **Exploratory Data Analysis (EDA):** Class distribution across A-E, Prompt Word Count Distribution, Correct Answer Count by Option, Prompt Length: Train vs Test, Correlation Heatmap, Max, Mean and Min len option Text sequence lengths.
- **Data Preprocessing:** duplicate & string cleaning, missing value handling, and structuring

pairs of (prompt, option).

4. Tokenization Strategy

- **Primary Tokenizer:** Tf-idf and AutoTokenizer associated with bert-base-uncased.
- **Implementation:** Built via Hugging Face AutoTokenizer using dynamic padding and truncation to max_length = 256. Sentence pairs were formatted as [CLS] Prompt [SEP] Option [SEP].

5. Modeling & Experimentation

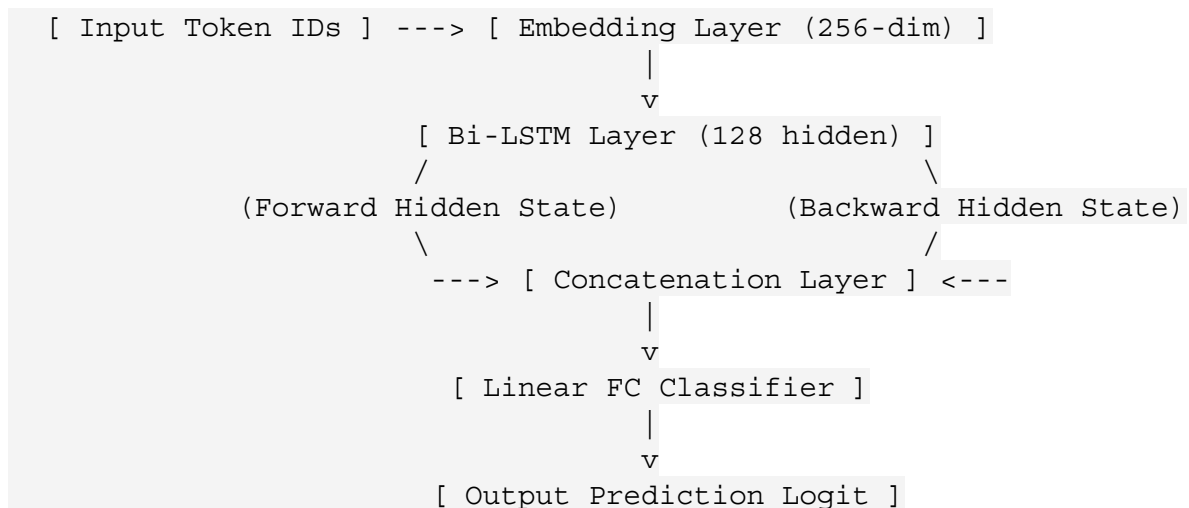
5.1 Model 1: Fine-Tuned Transformer Model (bert-base-uncased)

- **Architecture:** Pre-trained 12-layer bidirectional transformer encoder.
- **Hyperparameters:** batch_size: 16, epochs: 4, lr: 2e-5, max_len: 256.

5.2 Model 2: Custom Deep Learning Model (Bi-LSTM)

- **Architecture:** Bidirectional LSTM combined with an embedding layer and linear output projection head.
- **Hyperparameters:** vocab_size: 30,000, embedding_size: 256, hidden_size: 128, epochs: 13, lr: 3e-4, batch_size: 64.

Architectural Flow Diagram



5.3 Model 3: ML Model (TF-IDF + Logistic Regression)

- **Architecture:** 5-Fold Cross-Validated Logistic Regression on N-gram TF-IDF vectors.
- **Hyperparameters:** ngram_range: (1,3), tfidf_max_features: 60,000, C: 4.0, max_iter:

6. Performance & Comparative Analysis

Evaluation Metrics

Performance was measured using **MAP@3** (Primary Kaggle Metric) and **Validation Accuracy / Loss**.

Summary Results Table

Model Architecture	Key Configuration	Training Loss	Val Accuracy	Val MAP@3
Model 1: BERT-Base	lr: 2e-5, Epochs: 2, Batch: 16	0.0741	96.76%	0.9738
Model 2: BiLSTM	lr: 3e-4, Epochs: 13, Batch: 64	0.0027	96.0%	0.9617
Model 3: Logistic Regression	TF-IDF (1,3), 5-Fold CV, C=4	0.1433	99.97%	0.9974

Bert-base-uncased

Comprehensive Model Performance Visualization

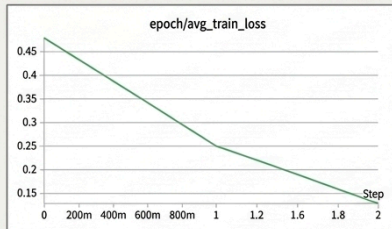


Figure 1a: Average Training Loss over Epochs

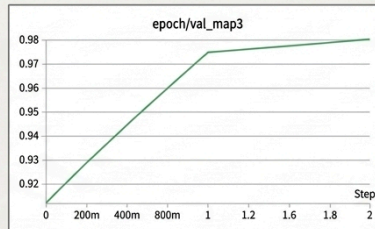


Figure 1b: Validation mAP@3 over Epochs

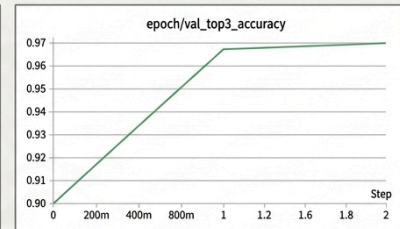


Figure 1c: Validation top3 accuracy over Epochs

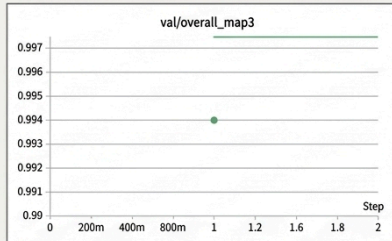


Figure 2c: Validation top3 accuracy over Epochs

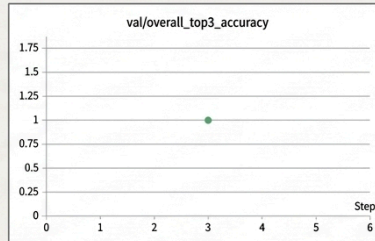


Figure 5e: Validation top3 accuracy over Epochs

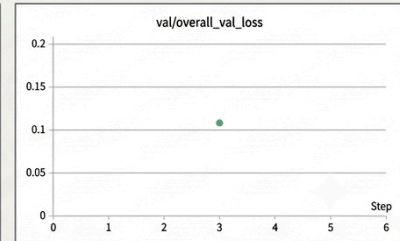
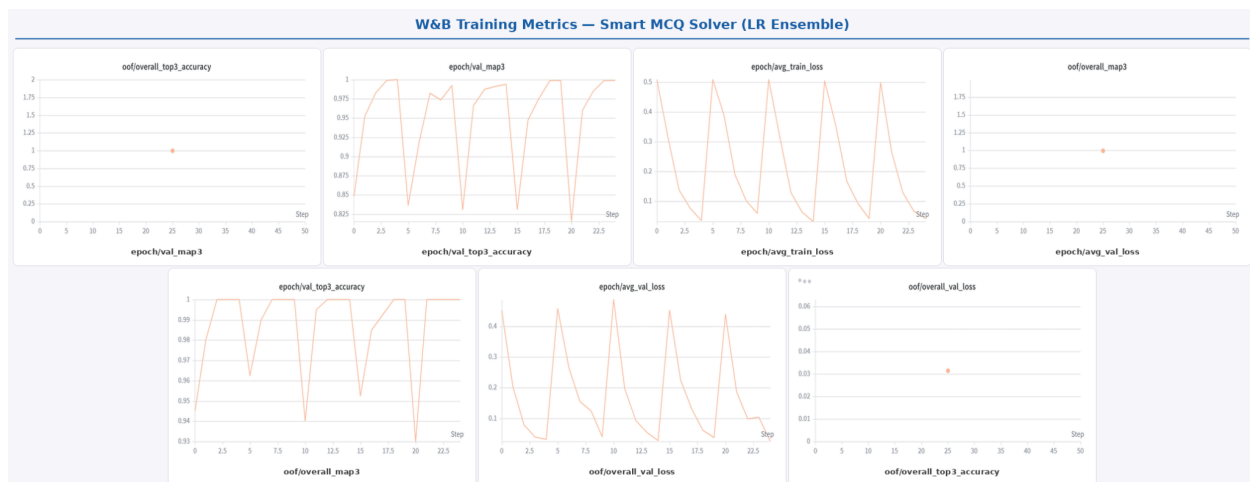


Figure 5f: Validation top3 accuracy over Epochs

Bilstm



LR

Comprehensive Model Performance Visualization

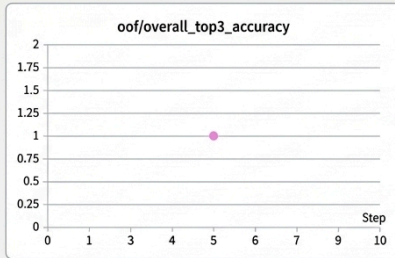


Figure 1: OOF Top-3 Accuracy

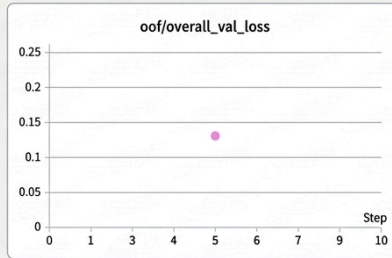


Figure 2: OOF Val Loss

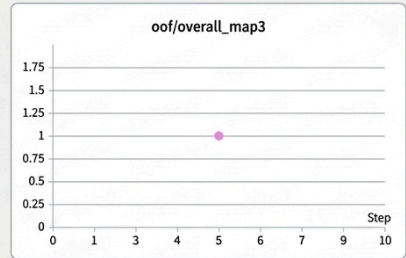


Figure 3: OOF Map3

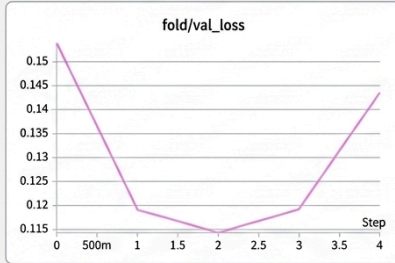


Figure 4: Fold Val Loss

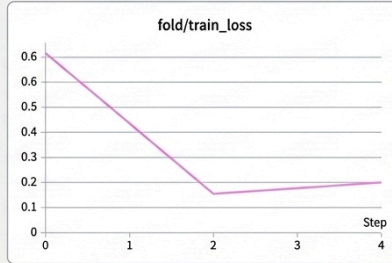


Figure 5: Fold Train Loss

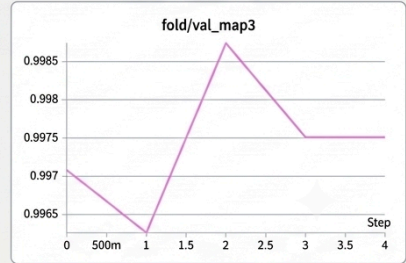


Figure 6: Fold Val Map3

7. Conclusion & Future Work

- **Key Learnings:** Fine-tuning Transformer models provides strong semantic generalizability, while custom Bi-LSTM models offer computationally lightweight yet competitive sequence classification performance.
- **Challenges:** Managing sequence padding, loss stabilization during multi-choice format conversion, and tuning learning rates for custom attention heads.
- **Future Scope:** Implementing ensemble methods combining BERT embeddings with Gradient Boosting (XGBoost/LightGBM) and exploring DeBERTa models for better ranking precision.

8. References

1. Pytorch Hugging face Documentation.
2. LLM